

Security and Risk in LLM Systems: Data, Leakage, and Adversarial Use

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Lecture 6

LLMs as Data Systems

LLMs must be understood as data-processing systems with security implications.

- ▶ They ingest and transform large volumes of data.
- ▶ They may reproduce patterns from training data.
- ▶ They blur boundaries between input and output.
- ▶ They function as probabilistic data interfaces.
- ▶ They introduce risks through normal usage.
- ▶ They must be treated as data systems.

From Failure to Security Risk

Failures in LLMs naturally become security concerns.

- ▶ Incorrect outputs may expose sensitive information.
- ▶ Hallucinations may misrepresent internal data.
- ▶ Misuse may cause unintended disclosure.
- ▶ Errors scale across many users.
- ▶ Risk accumulates over time.
- ▶ Reliability becomes governance.

The Data Problem

LLMs are trained on large datasets with limited transparency.

- ▶ Training sources are not fully disclosed.
- ▶ Data may include sensitive material.
- ▶ Provenance of information is unclear.
- ▶ Users cannot inspect training data.
- ▶ Boundaries of knowledge are unknown.
- ▶ Exposure risk exists without direct input.

Unknown Exposure Risk

The model may reproduce data without visibility.

- ▶ Memorized fragments may appear.
- ▶ Sensitive patterns may exist in weights.
- ▶ Outputs may reflect training data.
- ▶ Reproduction occurs under certain prompts.
- ▶ Risk is probabilistic.
- ▶ Verification is not possible.

Data Leakage Through Usage

User interaction creates additional exposure pathways.

- ▶ Conversations may be logged.
- ▶ Inputs may be retained.
- ▶ Users may share sensitive information.
- ▶ Internal processes may be exposed.
- ▶ Proprietary data may leave boundaries.
- ▶ Data lifecycle is unclear.

Input as Future Risk

User inputs may influence future system behavior.

- ▶ Inputs may persist across sessions.
- ▶ Data may enter training pipelines.
- ▶ Patterns may be generalized.
- ▶ Future outputs may reflect prior inputs.
- ▶ Boundaries between input and training blur.
- ▶ Users contribute to system knowledge.

Model Inversion Attacks

Attackers can extract data through systematic probing.

- ▶ Repeated queries reveal patterns.
- ▶ Sensitive data can be reconstructed.
- ▶ Outputs are refined iteratively.
- ▶ Memorization can be exploited.
- ▶ Statistical methods guide extraction.
- ▶ The model becomes a target.

Extraction Techniques

Specific techniques increase attack effectiveness.

- ▶ Contextual escalation narrows targets.
- ▶ Query saturation expands coverage.
- ▶ Low-temperature sampling stabilizes output.
- ▶ Prompt chaining refines results.
- ▶ Repetition amplifies signals.
- ▶ Iterative probing reconstructs data.

Model as Probeable System

The model behaves like a weakly protected database.

- ▶ It responds to structured queries.
- ▶ It lacks strict access control.
- ▶ It cannot detect user intent.
- ▶ Information may leak across sessions.
- ▶ Outputs expose latent knowledge.
- ▶ Reconstruction occurs over time.

Adversarial Use of LLMs

LLMs enable malicious applications at scale.

- ▶ Phishing and scam messages can be generated.
- ▶ Social engineering scripts can be created.
- ▶ Malware development may be assisted.
- ▶ Reconnaissance tasks can be automated.
- ▶ Attack content scales easily.
- ▶ Barriers to misuse are reduced.

Dual-Use Capability

LLMs support both beneficial and harmful uses.

- ▶ Security teams use models for analysis.
- ▶ Attackers use models for automation.
- ▶ Knowledge becomes widely accessible.
- ▶ Expertise is commoditized.
- ▶ Capabilities scale across users.
- ▶ Risk increases with accessibility.

Organizational Security Risk

Organizations face direct exposure through usage.

- ▶ Confidential data may be leaked.
- ▶ Internal workflows may be exposed.
- ▶ Credentials may be revealed.
- ▶ Data boundaries are difficult to enforce.
- ▶ Attack surface increases.
- ▶ Monitoring is incomplete.

Compliance Risk

LLMs introduce regulatory challenges.

- ▶ GDPR and CCPA violations may occur.
- ▶ Data retention policies are unclear.
- ▶ Data processing control is limited.
- ▶ Auditing is difficult.
- ▶ Unauthorized transfer may occur.
- ▶ Legal liability increases.

Trust Risk

Model behavior affects organizational trust.

- ▶ Outputs may be incorrect.
- ▶ Behavior may be inconsistent.
- ▶ Decisions are hard to explain.
- ▶ Users may overtrust outputs.
- ▶ Failures damage reputation.
- ▶ Confidence may degrade.

Enterprise Requirements

Organizations require strong governance properties.

- ▶ Systems must be auditable.
- ▶ Data flows must be controlled.
- ▶ Decisions must be traceable.
- ▶ Behavior must be reproducible.
- ▶ Accountability must be enforced.
- ▶ Risk must be measurable.

Fundamental Misalignment

LLMs conflict with enterprise expectations.

- ▶ Internal processes are opaque.
- ▶ Outputs are probabilistic.
- ▶ No inherent audit trail exists.
- ▶ Reproducibility is not guaranteed.
- ▶ Control mechanisms are limited.
- ▶ Explainability is weak.

Policy Mitigation

Policies reduce exposure risk.

- ▶ Restrict sensitive data entry.
- ▶ Define confidential information categories.
- ▶ Establish usage guidelines.
- ▶ Enforce system boundaries.
- ▶ Align with risk tolerance.
- ▶ Require compliance standards.

Training Mitigation

User education is essential.

- ▶ Train users on risks.
- ▶ Demonstrate failure cases.
- ▶ Encourage cautious input.
- ▶ Promote output verification.
- ▶ Build awareness of exposure.
- ▶ Develop literacy.

Technical Controls

Systems must enforce constraints programmatically.

- ▶ Disable data sharing when possible.
- ▶ Monitor interactions.
- ▶ Implement access controls.
- ▶ Filter sensitive data.
- ▶ Use secure environments.
- ▶ Limit external exposure.

Structural Insight

Risks arise from system design.

- ▶ Data exposure is inherent.
- ▶ Interaction creates risk.
- ▶ Adversaries exploit properties.
- ▶ Risks scale with usage.
- ▶ Problems persist across contexts.
- ▶ Mitigation must be systemic.

Synthesis

LLM risk is structural rather than incidental.

- ▶ It originates in training opacity.
- ▶ It emerges through interaction.
- ▶ It is amplified by adversaries.
- ▶ It cannot be eliminated.
- ▶ It must be managed continuously.
- ▶ It requires layered controls.

LLM Governance

LLMs must be treated as governed systems.

- ▶ They process sensitive data.
- ▶ They introduce measurable risk.
- ▶ They require explicit controls.
- ▶ They must meet enterprise standards.
- ▶ They need monitoring.
- ▶ They require oversight.

Discussion

LLMs must be integrated into enterprise governance frameworks.

- ▶ Policy frameworks are required.
- ▶ Architectural constraints must exist.
- ▶ Auditability is necessary.
- ▶ Compliance integration is required.
- ▶ Operational oversight is essential.
- ▶ Systems become infrastructure.

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